BALL MILL 5

TROUBLESHOOTING GUIDE

Fictional Industrial Equipment Documentation — Prototype Demonstration

Document No.	BM05-TG-001
Equipment Tag	BM-05
Equipment	Ball Mill 5
Model	FPM-BM5200-2C (Fictional)
Application	Cement Grinding
Revision	Rev. 01
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Status	SAMPLE / NON-OPERATIONAL

IMPORTANT: This guide is entirely fictional and is intended only for demonstrating a digital maintenance-documentation system. The fault codes, alarm limits, diagnostic steps and corrective actions must not be applied to real equipment without approved manufacturer and site procedures.

1. TROUBLESHOOTING PHILOSOPHY AND SAFETY

1.1 Purpose

This guide provides a structured fictional method for identifying common operating and maintenance symptoms associated with BM-05. It is organized around symptoms, possible causes, diagnostic checks and corrective actions.

1.2 Troubleshooting Principles

• Start with the simplest and safest possible explanation.

• Confirm the symptom using an actual reading, alarm or physical observation.

• Use trends and historical data when available.

• Do not bypass protective devices to restore production.

• Separate process problems from mechanical, electrical and instrumentation faults.

• Correct the root cause where possible rather than repeatedly resetting the same alarm.

1.3 Safety Requirements

WARNING: Troubleshooting around a ball mill may expose personnel to rotating equipment, high-voltage electrical systems, hot surfaces, stored energy, heavy components and process material. Any inspection requiring access to hazardous areas must follow the approved isolation and permit system.

1.4 Diagnostic Priority

Priority	Action
1	Protect people and equipment; stop if the condition is hazardous.
2	Record alarm, trip, time and operating conditions.
3	Check recent changes in feed, load, lubrication or auxiliary equipment.
4	Compare measurements with normal baseline trends.
5	Inspect the relevant system using approved methods.
6	Repair, test and document the outcome.

2. ALARM AND TRIP CODE DIRECTORY

2.1 Main Protection Codes

Code	Alarm / Trip	Severity	Initial Response
BM05-T01	Low lubrication pressure	Trip	Protect drive; investigate lubrication circuit.
BM05-T02	Main bearing high temperature	Trip	Stop/controlled shutdown and investigate.
BM05-T03	High mill vibration	Trip	Stop if required; inspect mechanical condition.
BM05-T04	Main motor overload	Trip	Investigate load and electrical protection.
BM05-T05	Emergency stop active	Trip	Do not reset until cause is confirmed safe.
BM05-T06	Gearbox oil high temperature	Alarm/Trip	Check oil flow, cooling and load.
BM05-T07	Lubrication filter differential pressure high	Alarm	Inspect filter condition.
BM05-T08	Motor bearing temperature high	Alarm/Trip	Investigate lubrication/cooling/mechanical condition.
BM05-T09	Downstream equipment unavailable	Permissive	Restore downstream readiness before starting.
BM05-T10	Mill start permissive not satisfied	Permissive	Identify open permissive in control system.

2.2 Alarm Handling

An alarm is an indication requiring attention; it is not automatically proof of a failed component. Verify the process value and sensor plausibility before replacing hardware.

2.3 Repeated Trips

Repeated trips should be treated as a fault condition. Record the sequence of events and investigate the initiating cause. Do not defeat or permanently bypass protection.

3. MILL WILL NOT START

3.1 Diagnostic Flow

Check	Possible Cause	Diagnostic Action	Corrective Direction
Emergency stop	E-stop active	Verify status at approved HMI/panel	Clear cause; reset under procedure.
Lubrication	No oil pressure	Check pump status, level and pressure indication	Restore lubrication system.
Downstream	Equipment unavailable	Check downstream permissive	Restore required readiness.
Motor protection	Trip active	Review protection indication	Authorized electrical investigation.
PLC permissive	Open permissive	Identify first open condition	Correct initiating condition.
Local/remote	Wrong control mode	Check selected control station	Select approved operating mode.
Instrumentation	Invalid signal	Compare indication with field condition	Inspect sensor/wiring.

3.2 Mill Starts Then Stops

Symptom	Likely Area	Checks
Stops immediately	Start permissive/protection	Review trip history and interlock sequence.
Stops after acceleration	Motor/load/mechanical	Review current, vibration and bearing values.
Stops after feed starts	Process/feed	Check feed rate, material condition and motor load.
Stops intermittently	Instrumentation/control	Review event timestamps and signal stability.

3.3 Recommended Record

Record start command time, motor current, lubrication pressure, bearing temperatures, active alarms, trip code, mill status and process feed condition.

4. HIGH VIBRATION

4.1 Symptom Assessment

High vibration should be investigated using trend data and measurements at consistent locations. Compare the current value with the established baseline rather than relying only on a single reading.

Possible Cause	Typical Indication	Diagnostic Check
Mechanical looseness	Increasing vibration with unusual noise	Inspect accessible fasteners/foundation under safe conditions.
Misalignment	Vibration associated with drive condition	Review alignment records and coupling condition.
Bearing deterioration	Temperature and vibration trend increase	Analyze vibration pattern and bearing temperature.
Liner/internal condition	Abnormal mill sound/vibration	Investigate during planned shutdown.
Unbalanced condition	Speed-related vibration	Review spectrum/trend where available.
Gearbox condition	Gear mesh-related vibration/noise	Review gearbox vibration and oil condition.

4.2 Corrective Strategy

• Verify the measurement instrument and sensor condition.

• Check whether vibration changed after maintenance, liner work or alignment work.

• Compare drive-end and non-drive-end measurements.

• Review gearbox and bearing temperature trends.

• Escalate significant or rapidly increasing vibration to engineering/maintenance.

4.3 Do Not

Do not continue operation solely because the vibration alarm can be reset. Do not enter the mill or remove guards while rotating equipment remains hazardous.

5. HIGH BEARING TEMPERATURE

Observation	Potential Cause	Diagnostic Approach
Both bearings rising	High load / oil temperature / cooling	Review load, oil supply temperature and cooling.
One bearing rising	Local lubrication or bearing issue	Compare oil flow and vibration; inspect local condition.
Temperature rises after startup	Lubrication/circulation issue	Confirm oil flow and pressure.
Temperature fluctuates	Sensor or intermittent flow	Compare redundant/independent readings.
Temperature rises with load	Process overload	Review feed and motor current.

5.1 Fictional Reference Values

Condition	Fictional Value
Normal main bearing temperature	35–65 °C
High alarm	70 °C
Trip	80 °C

5.2 Investigation Sequence

• Confirm the displayed temperature with an approved independent measurement where practical.

• Check lubrication pressure, flow and oil supply temperature.

• Inspect for leakage or restriction.

• Review vibration trend.

• Review recent maintenance and operating changes.

• If the temperature continues to rise, stop or reduce operation according to the approved protection philosophy and investigate.

5.3 Common Incorrect Diagnosis

A high temperature indication should not automatically be interpreted as a failed bearing. Sensor faults, poor lubrication, excessive load, cooling problems and mechanical misalignment can produce similar symptoms.

6. LUBRICATION AND GEARBOX PROBLEMS

6.1 Low Lubrication Pressure

Check	Possible Finding	Action
Reservoir level	Low oil level	Investigate loss/leakage and restore level using approved lubricant.
Pump	Pump not running	Check electrical/control status by authorized personnel.
Filter	High differential pressure	Inspect filter and contamination.
Piping	Restriction/leakage	Inspect under approved isolation.
Pressure sensor	Implausible reading	Verify indication and instrument condition.

6.2 High Gearbox Oil Temperature

Possible Cause	Diagnostic Check
Low oil level	Verify level and leakage.
Cooling problem	Check cooler performance and cooling medium.
High mechanical load	Review motor current and process conditions.
Oil degradation	Review oil analysis and service history.
Gearbox mechanical problem	Review vibration, noise and oil debris.

6.3 Abnormal Gearbox Noise

Document when the noise occurs, whether it changes with load, and whether it is accompanied by vibration or temperature changes. Review gearbox vibration data and oil condition before deciding on component replacement.

6.4 Lubrication Contamination

Water, particles or abnormal wear debris may indicate contamination or component deterioration. Take an approved oil sample and use trend analysis or laboratory results where available.

7. MOTOR, DRIVE AND PROCESS SYMPTOMS

7.1 Main Motor Overload

Possible Cause	Checks	Corrective Direction
Excessive feed	Review feed rate and process trend	Restore approved operating range.
Material buildup	Review process condition	Inspect during safe planned intervention.
Mechanical resistance	Review vibration, noise and temperatures	Mechanical investigation.
Electrical fault	Review protection and current balance	Authorized electrical testing.
Starting issue	Review acceleration/current history	Investigate motor/drive system.

7.2 Motor Does Not Reach Normal Speed

Check motor protection status, starting sequence, drive configuration where applicable, current behavior and mechanical load. Electrical diagnosis must be performed by authorized personnel using approved procedures.

7.3 Excessive Power Consumption

• Compare current feed rate with normal production conditions.

• Check material characteristics and mill loading.

• Review mill ventilation/process conditions where applicable.

• Inspect for abnormal mechanical resistance.

• Compare current and power trends with historical baseline.

7.4 Reduced Grinding Performance

Investigate feed characteristics, mill loading, grinding-media condition, liner condition, diaphragm condition, ventilation and process settings. Mechanical inspection should be planned rather than performed under unsafe operating conditions.

8. INSTRUMENTATION, CONTROL AND INTERMITTENT FAULTS

8.1 Implausible Sensor Reading

Symptom	Potential Cause	Check
Sudden temperature jump	Sensor/wiring issue	Compare with trend and independent measurement.
Pressure remains fixed	Sensor or signal fault	Compare with physical gauge/system behavior.
Vibration signal lost	Sensor/cable issue	Check diagnostic status and connections when safe.
Unstable reading	Loose connection/interference	Review signal quality and wiring.

8.2 Intermittent Trips

Intermittent faults are often difficult to diagnose because the equipment may appear normal after a reset. Capture the exact timestamp, active alarms, process conditions, sensor values and preceding maintenance activity.

Investigation Item	Record
Date/time	________________________
Trip code	________________________
Motor load	________________________
Oil pressure/temp.	________________________
Bearing temperature	________________________
Vibration	________________________
Feed condition	________________________
Recent maintenance	________________________

8.3 Control-System Checks

Review the alarm/event sequence and identify the first abnormal signal rather than the final trip generated by the control system. A downstream trip may be a consequence rather than the root cause.

8.4 Instrument Replacement

Replace sensors only after reasonable verification that the sensor or signal path is responsible. Record the removed component, replacement part, calibration status and post-replacement reading.

9. FAULT DIAGNOSIS RECORD AND QUICK REFERENCE

9.1 Fault Investigation Form

Field	Entry
Equipment	Ball Mill 5 / BM-05
Date & time	____________________________
Reported symptom	____________________________
Alarm / trip code	____________________________
Operating load	____________________________
Bearing temperature	____________________________
Oil pressure / temperature	____________________________
Vibration	____________________________
Initial suspected cause	____________________________
Inspection findings	____________________________
Root cause	____________________________
Corrective action	____________________________
Post-repair test	____________________________
Technician / supervisor	____________________________

9.2 Quick Diagnostic Matrix

Symptom	First Areas to Check
Won't start	Emergency stop, lubrication, permissives, motor protection, downstream readiness.
High vibration	Measurement validity, bearings, alignment, gearbox, foundation, internal condition.
High bearing temperature	Lubrication, cooling, load, vibration, sensor validity.
Low oil pressure	Oil level, pump, filter, restriction, leakage, pressure signal.
Gearbox overheating	Oil level, cooling, load, oil condition, vibration.
Motor overload	Feed/load, mechanical resistance, electrical protection.
Repeated trip	Event history, first abnormal signal, recent changes, instrumentation.

9.3 Escalation Criteria

Escalate when a fault involves repeated protective trips, rapidly increasing temperature or vibration, suspected structural damage, abnormal gearbox condition, electrical protection faults, fire, uncontrolled material release or any condition beyond the technician's approved authority.

FINAL DISCLAIMER: All codes, values, diagnoses and procedures in this ten-page guide are fictional. Use only approved manufacturer documentation, site procedures, risk assessments and competent engineering judgement when working on real equipment.

END OF DOCUMENT — BM05-TG-001 Rev. 01